Pizza Sales Analysis using SQL

1. Project Overview

This project focuses on analyzing pizza sales data using SQL to uncover key business insights related to revenue, customer ordering patterns, product performance, and sales trends. The goal of this analysis is to help improve business decision-making by identifying top-performing products, peak sales periods, and customer purchasing behavior.

2. Dataset Summary

Table Name: pizza_sales

Key Columns:

- Order Details: order_id, order_date, order_time
- Product Details**:** pizza_name, pizza_category, pizza_size
- Sales Metrics**:** quantity, total_price

Data Characteristics:

- Contains transactional-level sales data
- Each row represents a pizza item in an order
- Multiple rows can belong to the same order

3. Data Preparation (SQL-Based)

Before analysis, basic data validation and structuring were performed:

- Verified data consistency across columns
- Ensured correct data types for date and time fields
- Used SQL functions like DAYNAME(), HOUR(), and MONTH() for time-based analysis
- Used aggregate functions (SUM, COUNT, ROUND) for KPI calculations

4. Data Analysis using SQL (Business Insights)

A. KPI's

1. Total Revenue:

SELECT SUM(total_price) AS Total_Revenue FROM pizza_sales;

Results | Messages

	Total_Revenue
1	817860.05083847

2. Average Order Value

SELECT (SUM(total_price) / COUNT(DISTINCT order_id)) AS Avg_order_Value FROM pizza_sales;

Results | Messages

	Avg_order_Value
1	38.3072623343546

3. Total Pizzas Sold

SELECT SUM(quantity) AS Total_pizza_sold FROM pizza_sales;

Results | Messages

	Total_pizza_sold
1	49574

4. Total Orders

SELECT COUNT(DISTINCT order_id) AS Total_Orders FROM pizza_sales;

Results | Messages

	Total_Orders
1	21350

5. Average Pizzas Per Order

SELECT CAST(CAST(SUM(quantity) AS DECIMAL(10,2)) /

CAST(COUNT(DISTINCT order_id) AS DECIMAL(10,2)) AS DECIMAL(10,2))

AS Avg_Pizzas_per_order

FROM pizza_sales;

Results | Messages

	Avg_Pizzas_per_order
1	2.32

B. Daily Trend for Total Orders

SELECT DAYNAME(order_date) AS Order_Day, COUNT(DISTINCT order_id) AS Total_Orders

```
FROM pizza_sales

GROUP BY DAYNAME(order_date);
```

Output:

Results | Messages

	order_day	total_orders
1	Saturday	3158
2	Wednesday	3024
3	Monday	2794
4	Sunday	2624
5	Friday	3538
6	Thursday	3239
7	Tuesday	2973

C. Hourly Trend for Orders

```
SELECT HOUR(order_time) AS Order_Hours,

        COUNT(DISTINCT order_id) AS Total_Orders

FROM pizza_sales

GROUP BY HOUR(order_time)

ORDER BY HOUR(order_time);
```

Output

Results | Messages

	order_hours	total_orders
1	9	1
2	10	8
3	11	1231
4	12	2520
5	13	2455
6	14	1472
7	15	1468
8	16	1920
9	17	2336
10	18	2399
11	19	2009
12	20	1642
13	21	1198
14	22	663
15	23	28

D. % of Sales by Pizza Category

```
SELECT

    pizza_category,

    ROUND(SUM(total_price), 2) AS total_revenue,
```

```
    ROUND(SUM(total_price) * 100 / (SELECT SUM(total_price) FROM pizza_sales), 2) AS pct
FROM pizza_sales
GROUP BY pizza_category
ORDER BY total_revenue DESC;
```

Output

Results | Messages

	pizza_category	total_revenue	PCT
1	Classic	220053.10	26.91
2	Chicken	195919.50	23.96
3	Veggie	193690.45	23.68
4	Supreme	208197.00	25.46

E. % of Sales by Pizza Size

```
SELECT
    pizza_size,
    ROUND(SUM(total_price), 2) AS total_revenue,
    ROUND(SUM(total_price) * 100 / t.total_revenue, 2) AS pct
FROM pizza_sales,
     (SELECT SUM(total_price) AS total_revenue FROM pizza_sales) t
GROUP BY pizza_size, t.total_revenue
ORDER BY FIELD(pizza_size, 'S','M','L','XL','XXL');
```

Output

Results | Messages

	pizza_size	total_revenue	PCT
1	L	375318.70	45.89
2	M	249382.25	30.49
3	S	178076.50	21.77
4	XL	14076.00	1.72
5	XXL	1006.60	0.12

F. Total Pizzas Sold by Pizza Category

```
SELECT pizza_category, SUM(quantity) as Total_Quantity_Sold
FROM pizza_sales
```

```
WHERE MONTH(order_date) = 2

GROUP BY pizza_category

ORDER BY Total_Quantity_Sold DESC;
```

Output

Results | Messages

	pizza_category	Total_Quantity_Sold
1	Classic	14888
2	Supreme	11987
3	Veggie	11649
4	Chicken	11050

G. Top 5 Best Sellers by Total Pizzas Sold

```
SELECT pizza_name,

        SUM(quantity) AS Total_Pizza_Sold

FROM pizza_sales

GROUP BY pizza_name

ORDER BY Total_Pizza_Sold DESC

LIMIT 5;
```

Output

	pizza_name	Total_Pizza_Sold
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Pizza	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

H. Bottom 5 Best Sellers by Total Pizzas Sold

```
SELECT pizza_name,

        SUM(quantity) AS Total_Pizza_Sold
```

```
FROM pizza_sales

GROUP BY pizza_name

ORDER BY Total_Pizza_Sold ASC

LIMIT 5;
```

Output

Results | Messages

	pizza_name	Total_Pizza_Sold
1	The Brie Carre Pizza	490
2	The Mediterranean Pizza	934
3	The Calabrese Pizza	937
4	The Spinach Supreme Pizza	950
5	The Soppressata Pizza	961

5. Key Insights

- Certain days of the week generate higher order volumes, indicating peak business days
- Specific hours show significantly higher sales, highlighting peak ordering times
- A few pizza categories contribute the majority of total revenue
- Larger pizza sizes tend to generate higher revenue
- Top-selling pizzas drive a significant portion of sales
- Some pizzas consistently underperform and may require menu optimization

6. Tools & Technologies Used

- SQL (MySQL)
- Aggregate Functions (SUM, COUNT, ROUND)
- Date & Time Functions (DAYNAME, HOUR, MONTH)
- Grouping & Filtering (GROUP BY, ORDER BY, WHERE)
- Subqueries

7. Dashboard in Excel

Finally, we built an interactive dashboard in Excel to present insights visually.

8. Business Recommendations

- Optimize Menu: Focus on promoting top-selling pizzas and reconsider low-performing items
- Peak Hour Strategy: Increase staffing and marketing during high-demand hours
- Category Promotion: Boost sales of lower-performing categories through offers
- Size-Based Pricing Strategy: Promote high-revenue sizes (e.g., Large, XL)
- Seasonal/Monthly Analysis: Track monthly trends to adjust inventory and marketing

9. Conclusion

This project demonstrates how SQL can be effectively used to analyze transactional sales data and generate actionable business insights. The analysis helps in understanding customer behavior, optimizing product offerings, and improving overall sales performance.